

HAFIZ NAUMAN BAIG

PhD Research Scholar (Zoology / Fisheries & Aquaculture)

Department of Zoology, Faculty of Biological Sciences, Quaid-i-Azam University (QAU), Islamabad, Pakistan
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1. RESEARCH INTERESTS

Fish environmental and thermal stress physiology, sustainable intensive aquaculture, molecular growth regulation, oxidative stress biomarkers, fish histopathology, transcriptomic profiling, brain plasticity, and innovative dietary prophylactic strategies (e.g., natural antioxidants) to optimize fish health, welfare, and production in modern recirculating aquaculture systems (RAS).

2. EDUCATIONAL QUALIFICATIONS

Degree / Certificate	Institution / University	Duration / Year	Performance / Status
Ph.D. Zoology (Fisheries & Aquaculture)	Quaid-i-Azam University (QAU), Islamabad, Pakistan	Spring 2024 - Present	In Progress (Coursework: 18 Cr)
M.Phil. Zoology (Fisheries & Aquaculture)	The University of Lahore (UOL), Lahore, Pakistan	2021 - 2023	CGPA: 3.73 / 4.00 (First Division)
B.S. (4-Year) Zoology (Specialization: Fisheries)	University of the Punjab (PU), Lahore, Pakistan	2014 - 2018	First Division
F.Sc. (Pre-Medical)	Govt. Science College, Lahore, Pakistan	2011 - 2013	Grade A
Matriculation (Science)	Qazi Pilot High School, Lahore, Pakistan	2008 - 2010	Grade A+

3. M.PHIL. RESEARCH PROJECT

Title: "Physiological and Protective Role of Dietary Vitamin C Against ZnSO₄-Induced Toxicity in Nile Tilapia (*Oreochromis niloticus*)"

Supervisors: Miss Maham Mazher & Prof. Dr. Sikandar Hayat (IMBB, UOL)

Summary & Core Methodology: Investigated the oxidative stress, heavy-metal induced bioaccumulation, biochemical alterations, and correspondent tissue histopathology caused by sub-lethal zinc sulfate (ZnSO₄) exposure in juvenile Nile Tilapia. The study successfully demonstrated that therapeutic dietary ascorbic acid (Vitamin C) supplementation significantly ameliorates renal damage, muscle toxicity, and oxidative biomarkers. Key methodologies utilized included hematological profiling, colorimetric biochemical assays, antioxidant enzyme evaluation (SOD, CAT, TAC), and advanced liver/kidney microtomy and histopathology.

4. PROFESSIONAL RESEARCH & WORK EXPERIENCE

Fisheries Research Assistant

Department of Fisheries, Government of Punjab (Chakwal, Pakistan) | July 2019 - July 2020 (1 Year)

- Conducted extensive cage culture operations, feeding regime optimization, and biological sampling in major freshwater reservoirs.
- Monitored water quality parameters (Dissolved Oxygen, pH, Temperature, Ammonia, Hardness) to maintain optimal environmental standards.
- Assisted in aquatic disease surveillance, fish parasite identification, and general health management protocols of intensive systems.
- Collected, curated, and analyzed fish biomass and growth metrics to draft monthly operational productivity reports for administrative reviews.

5. PEER-REVIEWED SCIENTIFIC PUBLICATIONS

Baig, H. N., Mazher, M., & Hayat, S. "Biochemical and histopathological studies on the ameliorative effect of vitamin C on renal and muscle tissues of Nile tilapia (*Oreochromis niloticus*) exposed to zinc sulfate." (*Manuscript Under Review / Submitted*).

6. TECHNICAL & LABORATORY SKILLS

Technical Domain	Specific Methodologies & Core Competencies
Aquaculture & Physiology	Fish husbandry and RAS operations, water bio-chemistry monitoring, heavy-metal toxicity profiling, bio-assays, tissue micro-dissection, and animal health sampling.
Molecular Biology	Organelle/Tissue RNA extraction, cDNA synthesis, RT-qPCR gene expression profiling, colorimetric enzymatic assays, ELISA diagnostics, and biochemical histopathology.
Analytical & Soft Skills	Advanced statistical modeling in R-Studio (Linear Mixed Models [LMM], Zero-Inflated Beta [ZOIB] models), SPSS, MS Excel, scientific synopsis writing, and academic presentations.

7. PROFESSIONAL CERTIFICATIONS & TRAININGS

- **Invited Trainer & Presenter:** "Innovating for Tomorrow: Fish Health Management and Genetic Tools for Sustainable Fisheries" - International Symposium-cum-Workshop organized by Quaid-i-Azam University (QAU), Pakistan Fisheries Society, and HEC (Jan 2025).
- **Certificate of Completion:** "Ocean Governance: Sustainable Use of Ocean Resources" - International Ocean Institute (IOI) Ocean Academy (Sept - Oct 2025).
- **Government Professional Training:** "Extension, Conservation & Computer Skills" - Directorate of Fisheries R&T, Government of Punjab, Pakistan (Feb 2020).

8. ACADEMIC & PROFESSIONAL REFERENCES

1. Prof. Dr. Kiran Afshan

Primary Academic Referee & Chairperson
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Quaid-i-Azam University (QAU), Islamabad, Pakistan
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2. Prof. Dr. Sikandar Hayat

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